

1. Demo Introduction (30 seconds)

Say:

Hi Mahesh.

Today I will demonstrate our Axon Automation POC.

The objective of this solution is to eliminate the manual process of uploading evidence from DEMS into Axon.

Previously an agent had to manually log into Axon, locate the case, upload files, and verify the upload.

With our implementation, the agent only clicks “Move to Axon” in DEMS and the complete upload process happens automatically.

2. DEMS Demo

When you show DEMS UI:

Say:

Here we are in the DEMS application.

The agent selects the evidence and clicks Move to Axon.

This starts the synchronization process.

The user does not need to log in to Axon or perform any manual upload steps. The system handles everything automatically in the background.

3. Show Status

Say:

Once the request is submitted, a job is created.

The status moves through different stages such as Queued, In Progress, and finally Synced.

This allows the agent to track the upload without waiting for the entire upload process to complete.

4. Show Axon Portal Validation

After upload:

Say:

Now I will validate the result in Axon.

Here we can see that the evidence has been successfully uploaded to the correct case.

The file is available in Axon and linked with the same DEMS case number.

This confirms the end-to-end synchronization is successful.

5. Architecture Explanation

Show architecture image.

Say:

Let me explain the architecture at a high level.

Our design is asynchronous and scalable.

Each component has a specific responsibility.

Step 1 - DEMS Frontend

Say:

The process starts from the DEMS Frontend.

The agent clicks Move to Axon.

This creates a sync request.

Step 2 - API Gateway

Say:

The request is received by API Gateway.

API Gateway acts as the entry point for all synchronization requests.

It routes the request to our backend services.

Step 3 - Trigger Lambda

Say:

Trigger Lambda validates the request.

It creates a job record in DynamoDB.

It then places a message into SQS for processing.

Step 4 - DynamoDB

Say:

DynamoDB is our tracking database.

We store job status, success information, failure information, and duplicate detection information here.

This allows us to provide real-time status updates to DEMS users.

Step 5 - SQS

Most Important Question Area

Say:

SQS acts like a waiting queue.

Instead of processing uploads immediately, jobs are safely stored in the queue.

This prevents request loss and helps us handle multiple uploads reliably.

Even if ten agents click Move to Axon at the same time, all requests are safely queued and processed.

Why Two SQS Queues?

Say if somebody asks.

We use a primary queue and a dead-letter queue.

The primary queue processes normal upload requests.

If a message fails multiple times, it is automatically moved to the Dead Letter Queue.

This prevents endless retries and allows investigation of failed jobs.

6. EC2 Worker Explanation

This is your strongest section.

Say:

The heart of the solution is our Windows EC2 worker.

The worker continuously polls SQS for new jobs.

When a job arrives, it performs the actual Axon upload process.

Worker Steps

The worker:

- 1. Reads the job from SQS.*
- 2. Retrieves credentials from AWS Secrets Manager.*
- 3. Downloads the evidence file from S3.*
- 4. Launches Upload XT.*
- 5. Executes XTMT commands.*
- 6. Verifies upload success.*
- 7. Updates DynamoDB with the final status.*

7. Why S3?

Say:

S3 acts as the source repository where evidence files already exist.

The worker downloads files from S3 before starting the upload process.

8. Secrets Manager

Say:

Security was one of our major design considerations.

We do not hardcode usernames, passwords, or MFA secrets anywhere in the code.

All credentials are securely stored in AWS Secrets Manager.

The worker retrieves credentials only when required.

9. MFA Automation

This will definitely be asked.

Simple Version:

MFA was one of the biggest challenges.

Axon uses a time-based one-time password mechanism.

We securely store the MFA secret in AWS Secrets Manager.

During login, the worker automatically generates the current valid authentication code using the same industry-standard TOTP algorithm used by authenticator applications.

This allows fully automated login without requiring manual intervention.

10. Why We Use XTMT?

This is a very important question.

Say:

We evaluated browser-based uploads.

However, Axon's web portal has a file size limitation.

Axon provides an official desktop tool called Upload XT.

Upload XT contains an official command-line utility called XTMT.

Because XTMT is provided by Axon itself, it is more reliable and supports large evidence uploads.

Therefore we standardized on the XTMT approach.

11. Common XTMT Flags

If Mahesh asks.

Say:

We dynamically generate XTMT commands during runtime.

Important parameters include:

Case Number

Evidence File Path

Category

Description Metadata

Create Case If Not Exists

Auto Run Mode

These values are generated from the DEMS job data.

12. Why We Verify Uploads?

Say:

We do not simply assume upload success.

After XTMT execution, we read the XTMT progress logs.

Only after confirming success do we update DynamoDB and mark the job as Synced.

13. POC Achievements (End Slide)

Say:

Our POC successfully demonstrates:

- *DEMS to Axon automated synchronization*
- *Automated login and MFA handling*
- *Secure credential management through Secrets Manager*
- *Reliable queue-based processing using SQS*
- *Automated upload using Upload XT and XTMT*
- *Status tracking through DynamoDB*
- *End-to-end validation in Axon Portal*

Overall, we reduced a manual multi-step process to a single click experience for the agent.

Top 10 Questions Mahesh May Ask

Q1. Why use SQS?

Answer:

To decouple frontend from upload processing and ensure reliable job handling.

Q2. Why EC2?

Answer:

XTMT and Upload XT require a Windows environment and persistent execution.

Q3. Why not Lambda?

Answer:

Browser and desktop-tool based automation required capabilities that were better supported on EC2 during the POC.

Q4. Why Secrets Manager?

Answer:

Secure credential storage, no hardcoded passwords.

Q5. How MFA works?

Answer:

TOTP secret stored securely and codes generated dynamically.

Q6. Why XTMT?

Answer:

Official Axon tool, better for scripted uploads and large files.

Q7. How status updates work?

Answer:

DynamoDB stores job state and DEMS polls for updates.

Q8. What if upload fails?

Answer:

Status is marked Failed and the failed message can be investigated through the queue/error flow.

Q9. Duplicate protection?

Answer:

We maintain file tracking and duplicate detection logic before re-uploading.

Q10. Biggest achievement?

Answer:

Fully automated secure evidence transfer from DEMS to Axon with a one-click user experience.

 **My biggest advice for the demo: Speak slowly and simply. Don't try to sound like a senior architect. Just explain it as:**

“Agent clicks button → API → Lambda → SQS → EC2 → S3 + Secrets Manager → XTMT Upload → Axon → DynamoDB Status Update”

If you remember this single line, you can explain the entire architecture confidently. 